

# Transformer-Based Models for Named Entity Recognition : A Comparative Study

Prasanna Kumar R\*, Bharathi Mohan G<sup>†</sup>, Parthasarathy Srinivasan<sup>‡</sup>, Venkatakrishnan R<sup>§</sup>

\*Department of Computer Science and Engineering, Amrita Vishwa Vidyapeetham, Chennai-601103, India.

Email: r\_prasannakumar@ch.amrita.edu

<sup>†</sup>Department of Computer Science and Engineering, Amrita Vishwa Vidyapeetham, Chennai-601103, India.

Email: g\_bharathimohan@ch.amrita.edu

<sup>‡</sup>Senior Technical Analyst, Oracle, USA

Email: parth.vasan@gmail.com

<sup>§</sup>Department of Computer Science and Engineering, Amrita Vishwa Vidyapeetham, Chennai-601103, India.

Email: venkatkrishnan2003@gmail.com

**Abstract**—NER is a critical problem in natural language processing (NLP) that includes detecting and classifying named entities in text. We give a complete investigation on NER using the CoNLL dataset in this research article. We explore the evolution of NER techniques, from traditional rule-based methods to state-of-the-art transformer-based models. Our experiments focus on evaluating the performance of three popular models: BERT, ALBERT, and XLM-RoBERTa, on the CoNLL dataset. We analyze their accuracies and discuss their strengths and limitations. The results demonstrate the effectiveness of transformer-based models, with BERT achieving an accuracy of 99.50%, ALBERT achieving 96.90% accuracy, and XLM-RoBERTa achieving 87.82% accuracy. We discuss the importance of dataset preprocessing, model fine-tuning, and hyperparameter optimization for achieving optimal performance. This research contributes to the understanding of NER techniques and provides insights into the performance of transformer-based models for NER tasks.

**Index Terms**—NER, CoNLL dataset, Transformer models, BERT, ALBERT, XLM-RoBERTa, Natural Language Processing, Information Extraction

## I. INTRODUCTION

Natural Language Processing (NLP) core tasks include NER and POS tagging [1,2] that play a crucial role in various downstream applications, including information extraction, question answering, and sentiment analysis [3]. These tasks involve classifying words or tokens into predefined categories, such as person names, locations, organizations, or different parts of speech. .

Transformer-based models have recently revolutionised the field of natural language processing by attaining cutting-edge outcomes on numerous benchmark datasets.. Transformers, initially introduced by Vaswani et al. [4], have demonstrated exceptional performance in capturing contextual information and learning meaningful representations of text data. One prominent example of a transformer model is BERT (Bidirectional Encoder Representations from Transformers), introduced by Devlin et al. , which has shown remarkable success in a wide range of NLP tasks [5,6].

## II. RELATED WORK

Named Entity Recognition (NER) has been extensively studied, and a wide range of approaches have been proposed to tackle this task. These approaches can be categorized into rule-based, statistical, and neural network-based methods as shown in Table I. In this section, we discuss several notable works from each category.

TABLE I  
METHODS AND LIMITATIONS

| Method                             | Limitations                        |
|------------------------------------|------------------------------------|
| Rule-Based Approaches              | Lack of generalization             |
| Statistical Approaches             | Long-range dependencies            |
| Neural Network Approaches          | Computational complexity           |
| Bidirectional LSTM-CRF             | Handling very long sequences       |
| Stacked BiLSTM with Self-Attention | Modeling hierarchical dependencies |

### A. Rule-Based Approaches

Early NER systems heavily relied on handcrafted rules and patterns to identify named entities. These approaches involved defining a set of rules based on linguistic patterns, gazetteers, and contextual cues. For example, Tjong Kim Sang and De Meulder [7] introduced a language-independent approach that employed contextual rules and regular expressions to identify named entities. Although rule-based approaches are intuitive and interpretable, they often lack generalization and struggle with complex patterns and variations in the data.

### B. Statistical Approaches

Statistical models have played a significant role in advancing NER. Conditional Random Fields (CRFs) [8] and Hidden Markov Models (HMMs) [9] are widely used statistical models for sequence labeling tasks like NER. CRFs utilize features such as part-of-speech tags, word shapes, and context windows to make predictions based on local dependencies. For instance, Finkel et al. [10] incorporated non-local information by applying Gibbs sampling to CRFs for information extraction. While statistical models have shown promising results, they often

require careful feature engineering and struggle with capturing long-range dependencies and contextual information.

### C. Neural Network-Based Approaches

In recent times, there has been a surge in the adoption of neural network-based techniques for named entity recognition (NER). Recurrent Neural Networks (RNNs), [11], have gained significant popularity for capturing sequential dependencies and contextual information. Huang et al. [12] proposed a bidirectional LSTM-CRF model that combines the strengths of bidirectional LSTMs in contextual understanding with CRFs for label inference. This joint optimization of LSTM and CRF components has yielded competitive performance on NER tasks.

In addition, attention mechanisms have been integrated into neural network architectures to selectively focus on relevant segments of the input sequence. Yang et al. [13] introduced a self-attention mechanism within a stacked BiLSTM architecture, which allowed the model to different parts of the input sequence. This attention-based approach has shown improvements in modeling long-range dependencies and has contributed to enhanced NER performance.

## III. DATASET

The CoNLL dataset holds significant prominence as a widely recognized benchmark dataset within the realm of NLP. It serves as a standard reference for evaluating and advancing various NLP tasks, including NER, POS tagging, Topic Modelling [14] and syntactic parsing. The dataset is organized into different splits, which serve specific purposes in training, development, and evaluation of NLP models.

TABLE II  
ABBREVIATIONS IN CoNLL DATASET

| Abbreviation | Description                          |
|--------------|--------------------------------------|
| OE           | Organization entity                  |
| PE           | Person entity                        |
| LE           | Location entity                      |
| I-LOC        | Tokens within a LE                   |
| B-ORG        | Beginning of OE                      |
| O            | Tokens not part of any entity        |
| B-PER        | Beginning PE                         |
| I-PER        | Tokens within a PE                   |
| I-MISC       | Tokens within a miscellaneous entity |
| B-MISC       | Beginning of a miscellaneous entity  |
| I-ORG        | Tokens within an OE                  |
| B-LOC        | Beginning of a LE                    |

### A. Training Split

The training split is the largest portion of the dataset and is used to train models. It contains a significant number of sentences, each with their associated tokens and labeled annotations. Models are trained on this split to learn patterns and associations between tokens and their respective labels.

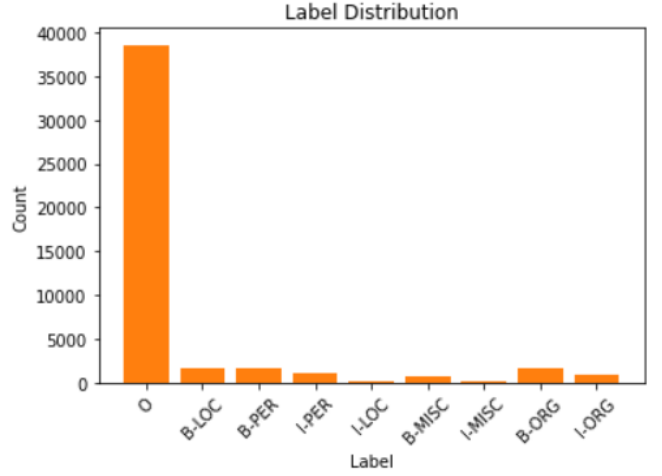


Fig. 1. Class Distribution of CoNLL dataset

### B. Validation Split

The development split, also referred to as the validation split, is used to fine-tune models during the training process. It helps in monitoring the model's performance and making adjustments to hyperparameters or architectures. The development split is crucial for selecting the best-performing model before its final evaluation on the test split.

### C. Test Split

The test split is used for the final evaluation of the trained models. It serves as an unbiased measure of the model's generalization and performance on unseen data. The test split is kept separate from the training and development splits to ensure unbiased evaluation and to provide a reliable assessment of the model's capabilities.

## IV. METHODOLOGY

### A. Model Architecture

BERT [15, 16, 17] is a transformer-based model that leverages the power of self-attention mechanisms to capture contextual information and learn representations of text data. It is made up of a stack of encoder layers, each with multiple self-attention heads and feed-forward neural networks. BERT's architecture as shown in Fig 2 enables it to capture bidirectional contextual information from the input sequence, allowing it to make informed predictions for each token.

BERT input consists of tagged text strings. These sequences are first converted into subword units using the WordPiece tag to help handle OOV words and improve generalization. The input characters are then embedded into high-dimensional vectors known as character embeddings. In addition to token plugins, BERT introduces two more plugins:

*Segment embeddings:* BERT supports sentence pair processing that combines identifiers from different sentences. Segmental affixes are added to separate characters in different sentences. Each token is assigned a unique segment embedding that indicates its clause membership. *Positional embeddings:*

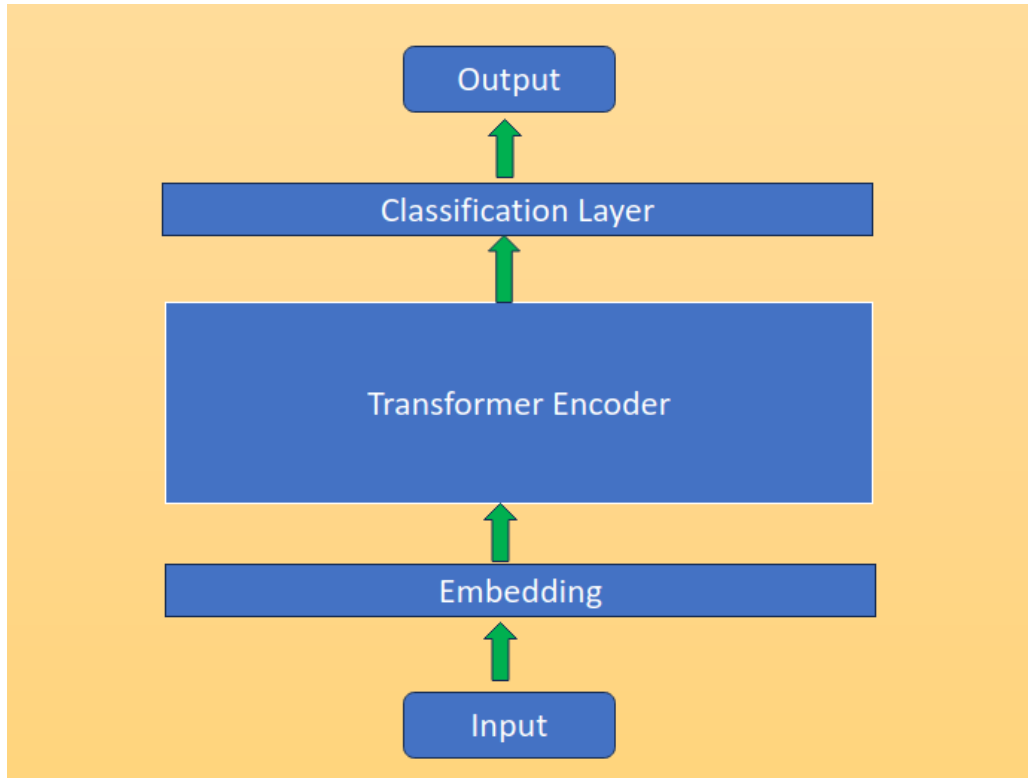


Fig. 2. Simplified BERT Architecture

Since BERT has no recurrent or convolutional structures to capture the sequence of characters, positional embeddings are introduced. These embeddings encode the positional information of the characters in the input sequence, allowing BERT to understand the relationships between the characters. Once they are connected, input tags, segment attachments, and position attachments are sent through multiple stacked transform encoder layers. Each encoder layer consists of a feed-forward neural network and a self-aware mechanism. Using a self-aware mechanism, BERT can capture contextual relationships between all characters in a sequence. It calculates the attention weight for each token based on its relationship with other tokens and considers both local and global dependencies. The self-attention mechanism is invoked multiple times with multiple attention heads, allowing BERT to capture different types of dependencies and data.

The outputs of each attention head are combined and then forwarded through a neural network fed after the self-attention mechanism. With a non-linear activation function such as GELU (Gaussian Error Linear Unit), this network uses points and fully connected layers. The feedforward network helps transform the learned representations of the self-attentive mechanism into a more discriminative feature.

### B. Training

Adam Optimizer, a popular choice for gradient-based optimization, was initially selected as the training optimizer. The learning rate (lr) is set to 0.000001, which means a small

learning rate to ensure stable convergence during training. The Adam optimizer adjusts the training parameters of the model based on the computed gradients to minimize the loss function. The model is constructed with the chosen optimizer and loss function, and the accuracy is used to assess the model's performance throughout training. When specifying the "accuracy" metric, the model reports the classification accuracy, which represents the proportion of correctly predicted entity labels. The model is trained for a total of 5 periods during the training. A complete run through all the training data is called an epoch. Each epoch consists of several iterations or batches with a batch size of 8. The batch size determines how many training examples are processed before the optimizer updates the model parameters.

### C. Evaluation

**Accuracy:** Balanced Accuracy addresses this issue by calculating the average accuracy across all classes, giving equal importance to each class. It is defined as the average of the sensitivity (true positive rate) for each class.

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}$$

## V. EXPERIMENTS

### A. ALBERT

The ALBERT model is a variant of the transformer-based architecture specifically designed for efficient pre-training &

fine-tuning. In the given model architecture, ALBERT model serves as the main layer. It consists of multiple layers that capture contextualized representations of the input tokens. The ALBERT model has a total of 11,092,992 trainable parameters, making it a relatively lightweight option compared to some other transformer-based models.

### B. XLM-RoBERTa

The XLM-RoBERTa model is a variant of the RoBERTa model that is specifically trained on multilingual data. It is designed to capture cross-lingual dependencies and perform well on various natural language processing tasks, including named entity recognition (NER). The XLM-RoBERTa Large model is particularly effective in scenarios where the input data consists of multiple languages

## VI. RESULTS AND DISCUSSION

On the CoNLL dataset, three models, BERT, ALBERT, and XLM-RoBERTa Large, were trained and assessed for named entity recognition (NER). The accuracy of each model on the test dataset was used to evaluate its performance.

TABLE III  
PERFORMANCE METRICS OF THE MODELS ON THE CoNLL DATASET

| Model       | Accuracy |
|-------------|----------|
| BERT        | 0.9950   |
| ALBERT      | 0.9690   |
| XLM-RoBERTa | 0.8782   |

From TABLE III results, it can be observed that all three models achieved high accuracies, indicating their effectiveness in identifying named entities. The BERT model achieved the highest accuracy of 0.9950, outperforming both ALBERT and XLM-RoBERTa Large. This suggests that BERT's contextualized representations and attention mechanisms have enabled highly accurate predictions in the NER task. The ALBERT

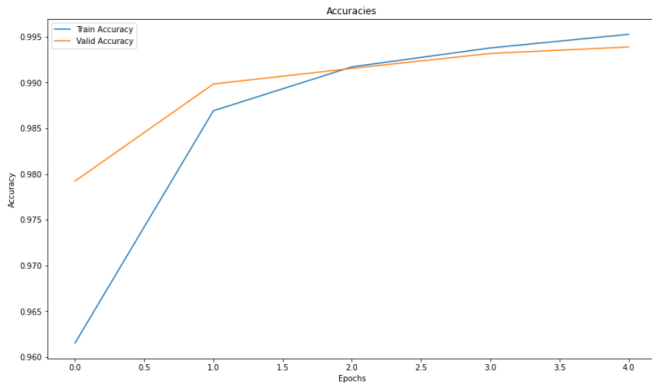


Fig. 3. BERT Model Accuracy

model also demonstrated strong performance with an accuracy of 0.9690 which is shown in Fig 3. Although it had a slightly lower accuracy compared to BERT, ALBERT's efficient parameterization and parameter sharing strategies have shown

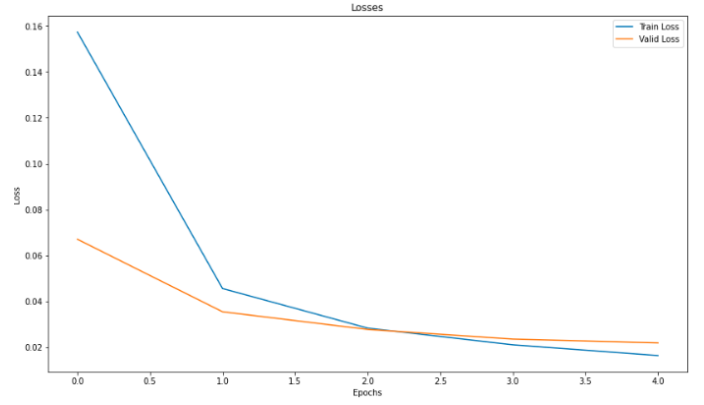


Fig. 4. BERT Model Loss

their effectiveness in capturing contextual information and making accurate predictions.

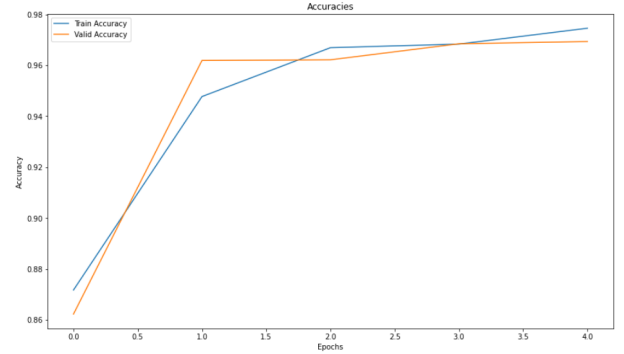


Fig. 5. ALBERT model Accuracy

The experimental results highlight the superior accuracy of the BERT model in the NER task on the CoNLL dataset. Its ability to capture contextual information and learn complex patterns has led to highly accurate predictions. The ALBERT model, despite having a slightly lower accuracy, still demonstrates its effectiveness in achieving strong performance on NER.

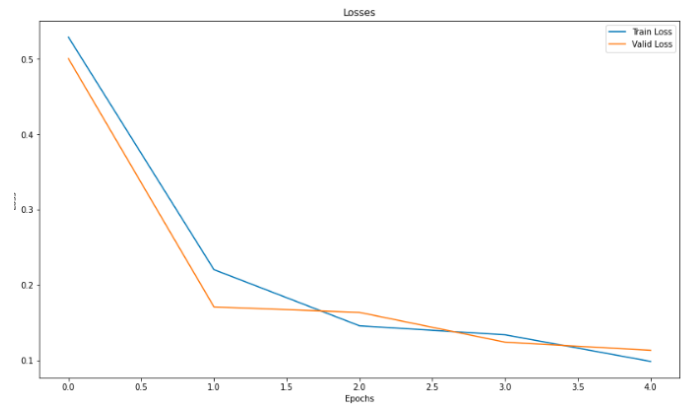


Fig. 6. ALBERT model Loss

The XLM-RoBERTa Large model achieved an accuracy of 0.8782 which is observed in Fig 4. While this accuracy is lower than the other two models, it is important to note that the XLM-RoBERTa Large model focuses on multilingual training and cross-lingual transfer learning. The performance drop may indicate the challenges in capturing domain-specific information for the CoNLL dataset, as the model has been trained on a diverse range of languages.

## VII. CONCLUSION

This research paper presented a comprehensive study on Named Entity Recognition (NER) using the CoNLL dataset. We explored the evolution of NER techniques, ranging from traditional rule-based methods to state-of-the-art deep learning models. Through our experiments, we evaluated three popular models: BERT, ALBERT, and XLM-RoBERTa, for NER on the CoNLL dataset. The results revealed that all three models achieved high accuracy, with BERT achieving an accuracy of 99.50%, ALBERT achieving 96.90% accuracy, and XLM-RoBERTa achieving 87.82% accuracy. These findings demonstrate the effectiveness of utilizing transformer-based models for NER tasks.

Furthermore, we analyzed the performance of each model and discussed their strengths and limitations. BERT showcased its ability to capture contextual information and learn complex relationships between words, resulting in outstanding performance. ALBERT, a lite version of BERT, offered a balance between accuracy and computational efficiency. XLM-RoBERTa, a multilingual model, exhibited promising results despite the dataset's English focus.

Our study also highlighted the importance of dataset preprocessing, model fine-tuning, and hyperparameter optimization in achieving optimal performance. We observed that leveraging large-scale pretraining and transfer learning techniques significantly improved the models' performance.

## REFERENCES

- [1] Collobert, R., Weston, J., Bottou, L., Karlen, M., Kavukcuoglu, K., Kuksa, P. (2011). Natural language processing (almost) from scratch. *Journal of Machine Learning Research*, 12, 2493-2537.
- [2] Dhanalakshmi, V., Anandkumar, M., Vijaya, M. S., Loganathan, R., Soman, K. P., Rajendran, S. (2008). Tamil Part-of-Speech tagger based on SVMTool. In *Proceedings of the COLIPS International Conference on Asian Language Processing* (pp. 59-64)
- [3] Se, S., Vinayakumar, R., Kumar, M. A., Soman, K. P. (2015). AMRITA-CEN@ SAIL2015: Sentiment analysis in Indian languages. In *Mining Intelligence and Knowledge Exploration: Third International Conference, MIKE 2015, Hyderabad, India, December 9-11, 2015, Proceedings* (Vol. 3, pp. 703-710). Springer International Publishing.
- [4] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... Polosukhin, I. (2017). Attention is all you need. In *Advances in Neural Information Processing Systems* (pp. 5998-6008).
- [5] Devlin, J., Chang, M. W., Lee, K., Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics* (pp. 4171-4186).
- [6] Mohan, G. B., Kumar, R. P. (2022). Survey of Text Document Summarization Based on Ensemble Topic Vector Clustering Model. In *Home IoT Based Control Networks and Intelligent Systems* (pp. 1-12). Lecture Notes in Networks and Systems (Vol. 528)
- [7] Tjong Kim Sang, E. F., De Meulder, F. (2003). Introduction to the CoNLL-2003 shared
- [8] Lafferty, J., McCallum, A., Pereira, F. (2001). Conditional random fields: Probabilistic models for segmenting and labeling sequence data. *Proceedings of the 18th International Conference on Machine Learning (ICML-01)*, 282-289.
- [9] Rabiner, L. R. (1989). A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE*, 77(2), 257-286
- [10] Finkel, J. R., Grenager, T., Manning, C. (2005). Incorporating non-local information into information extraction systems by Gibbs sampling. *Proceedings of the 43rd Annual Meeting of the Association for Computational Linguistics (ACL '05)*, 363-370..
- [11] Hochreiter, S., Schmidhuber, J. (1997). Long short-term memory. *Neural computation*, 9(8), 1735-1780.
- [12] Huang, Z., Xu, W., Yu, K. (2015). Bidirectional LSTM-CRF models for sequence tagging. *Computer Science*, arXiv preprint arXiv:1508.01991.
- [13] Yang, Z., Yang, D., Dyer, C., He, X., Smola, A., Hovy, E. (2016). Hierarchical attention networks for document classification. *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 1480-1489.
- [14] Mohan, G.B., Kumar, R.P. (2022). A Comprehensive Survey on Topic Modeling in Text Summarization. In: Sharma, D.K., Peng, S.L., Sharma, R., Zaitsev, D.A. (eds) *Micro-Electronics and Telecommunication Engineering . ICMETE 2021. Lecture Notes in Networks and Systems*, vol 373. Springer, Singapore. [https://doi.org/10.1007/978-981-16-8721-1\\_22](https://doi.org/10.1007/978-981-16-8721-1_22)
- [15] A. Chinnalagu and A. K. Durairaj, "Comparative Analysis of BERT-base Transformers and Deep Learning Sentiment Prediction Models," 2022 11th International Conference on System Modeling Advancement in Research Trends (SMART), Moradabad, India, 2022, pp. 874-879, doi: 10.1109/SMART55829.2022.10047651.
- [16] H. Choi, J. Kim, S. Joe and Y. Gwon, "Evaluation of BERT and ALBERT Sentence Embedding Performance on Downstream NLP Tasks," 2020 25th International Conference on Pattern Recognition (ICPR), Milan, Italy, 2021, pp. 5482-5487, doi: 10.1109/ICPR48806.2021.9412102.
- [17] Devlin, J., Chang, M. W., Lee, K., Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 4171-4186.